

Control work 1 — expressions and equations

Directed numbers, algebraic expressions and one-step and two-step equations.

LESSON 19

1 × 40 MIN

MATEMATIKA 6, NAZORAT ISHI 1

S7 1–2 REVIEW

LESSON CLOCK

2'

28'

8'

2'

Instructions	2 min
The paper	28 min
Answers and diagnosis	8 min
What comes next	2 min

LEARNING OBJECTIVES

- Work with the sign rules accurately under time.
- Translate between words and algebra.
- Simplify and evaluate an expression.
- Solve an equation and check the root.

TEXTBOOK

National	pp. 9–50
Cambridge	Stage 7, pp. 8–38
Workbook	Control paper 1

01

Explanation

The paper — 20 marks, 28 minutes

Q	TASK	MARKS	FROM
1	$-3 + 8$, $4 - 9$ and $5 - (-6)$	3	L2–4
2	$(-7) \cdot 4$, $(-36) \div (-9)$ and $(-3)^2$	3	L5–7
3	Write “four times a number, then subtract 3” and “subtract 3 from a number, then multiply by 4”	3	L8–10
4	Simplify $6x + 2y - 4x + 3y$	3	L11–12
5	Find $x^2 - 2x$ at $x = -3$	3	L13–15
6	Solve $4x - 5 = 19$ and check	3	L16–18
7	Solve $3(x + 2) = 5x$	2	L16–18

THE ANSWERS

5, -5, 11; -28, 4, 9; $4x - 3$ and $4(x - 3)$; $2x + 5y$; 15; $x = 6$; $x = 3$.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
double minus not applied	$5 - (-6) = -1$	11
sign rule reversed	$(-36) \div (-9) = -4$	4
$(-3)^2$	-9	9
bracket omitted in a translation	$4x - 3$	$4(x - 3)$
unlike terms collected	$6x + 2y = 8xy$	$6x + 2y$
negative substituted without brackets	$-3^2 = -9$	$(-3)^2 = 9$
no check	a root and nothing else	substitute into the original

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

What comes next

IF YOU LOST MARKS ON	REVISE BEFORE
Q1–Q2	the next Cambridge insert — inequalities, lesson 20
Q3–Q4	ratio, lesson 42, where expressions return
Q5	percentages, lesson 59, where formulae are evaluated
Q6–Q7	every word problem for the rest of the year

LOOKING FORWARD

The next block leaves algebra for a while and returns to fractions — but every one of the equations above will be needed again in Quarter III, when speed and distance are found from formulae.

02 Worked examples

EXAMPLE 1 Model answer, Q5: find $x^2 - 2x$ at $x = -3$.

① Substitute in brackets: $(-3)^2 - 2(-3)$.

② $(-3)^2 = 9$

③ $-2 \cdot (-3) = +6$
Two minuses.

④ $9 + 6 = 15$

ANSWER

15

EXAMPLE 2 Model answer, Q6: solve $4x - 5 = 19$.

- ① Add 5: $4x = 24$.
- ② Divide by 4: $x = 6$.
- ③ Check: $4 \cdot 6 - 5 = 19$ ✓

ANSWER

$x = 6$

EXAMPLE 3 Model answer, Q7: solve $3(x + 2) = 5x$.

- ① Expand: $3x + 6 = 5x$.
- ② Subtract $3x$: $6 = 2x$.
- ③ $x = 3$
Check: $15 = 15$ ✓

ANSWER

$x = 3$

03 Interactive model

Mark Q1 and Q2 first and return them within the lesson; sign errors corrected the same day rarely come back.

The block in eight questions

$5 - (-6)$ equals:

-1

11

1

-11

$(-7) \cdot 4$ equals:

28

-28

-3

3

$(-36) \div (-9)$ equals:

4

-4

-45

45

$(-3)^2$ equals:

9

-9

6

-6

“Subtract 3 from a number, then multiply by 4”:

$4x - 3$

$4(x - 3)$

$4x + 3$

$3 - 4x$

$6x + 2y - 4x + 3y$ equals:

$2x + 5y$

$10x + 5y$

$2x - y$

$7xy$

$x^2 - 2x$ at $x = -3$ is:

3

15

-15

9

$4x - 5 = 19$ gives:

$$x = 6$$

$$x = 3.5$$

$$x = 24$$

$$x = 4$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Integer	Butun son	Целое число
Expression	Ifoda	Выражение
Like terms	O'xshash hadlar	Подобные члены
To substitute	O'rniga qo'yish	Подставить
Equation	Tenglama	Уравнение
Root	Ildiz	Корень
Check	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 tests:

multiplying integers

adding and subtracting integers

equations

expressions

Q3 loses marks most often through:

the numbers

the missing bracket

the letters

the sign

Q4 answer is:

 $2x + 5y$ $10x + 5y$ $7xy$ $2xy$

Q5 needs the substitution written:

without brackets

in brackets

as a decimal

as a fraction

Q6 must end with:

the root

the root and a check

a graph

a table

Q7 begins by:

dividing

expanding the bracket

squaring

checking

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $-3 + 8$

ANSWER 5

2. $4 - 9$

ANSWER -5

3. $5 - (-6)$

ANSWER 11

4. $(-7) \cdot 4$

ANSWER -28

5. $(-36) \div (-9)$

ANSWER 4

6. $(-3)^2$

ANSWER 9

7. $6x + 2y - 4x + 3y$

ANSWER $2x + 5y$ **Medium · the standard question**

7 PROBLEMS

1. “Four times a number, then subtract 3”

ANSWER $4x - 3$

2. “Subtract 3 from a number, then multiply by 4”

ANSWER $4(x - 3)$

3. $x^2 - 2x$ at $x = -3$

ANSWER 15

4. $4x - 5 = 19$

ANSWER $x = 6$

5. $3(x + 2) = 5x$

ANSWER $x = 3$

6. $2x + 7 = x + 12$

ANSWER $x = 5$

7. $(-2)^3$

ANSWER -8

Hard · stretch and reason

7 PROBLEMS

1. $-4 - (-9) + (-2)$

ANSWER 3

2. $(-2)(-3)(-1)$

ANSWER -6

3. $2x^2 - x$ at $x = -2$

ANSWER 10

4. $5(x - 1) = 3(x + 3)$

ANSWER $x = 7$

5. Simplify $4(a + 2b) - 3a$

ANSWER $a + 8b$

6. A number trebled and decreased by 4 is 17

ANSWER 7

7. Why must -3 be substituted in brackets?

ANSWER Otherwise the square loses its sign

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on in full, from the first line.

1. Rewrite in full every question on which you lost a mark.
2. Compute $-8 + 15$, $3 - 11$ and $7 - (-4)$.
3. Compute $(-5) \cdot 6$ and $(-48) \div (-6)$.
4. Simplify $8a + 3b - 5a + b$ and find its value at $a = 2$, $b = -1$.
5. Solve $5x - 4 = 26$ and check your root.